BUILD A GREENER WORLD

CREATE A BETTER LIFE

Power Atlantic 5MWh Battery Container

User Manual

(Preliminary Version)

PREPARED BY:

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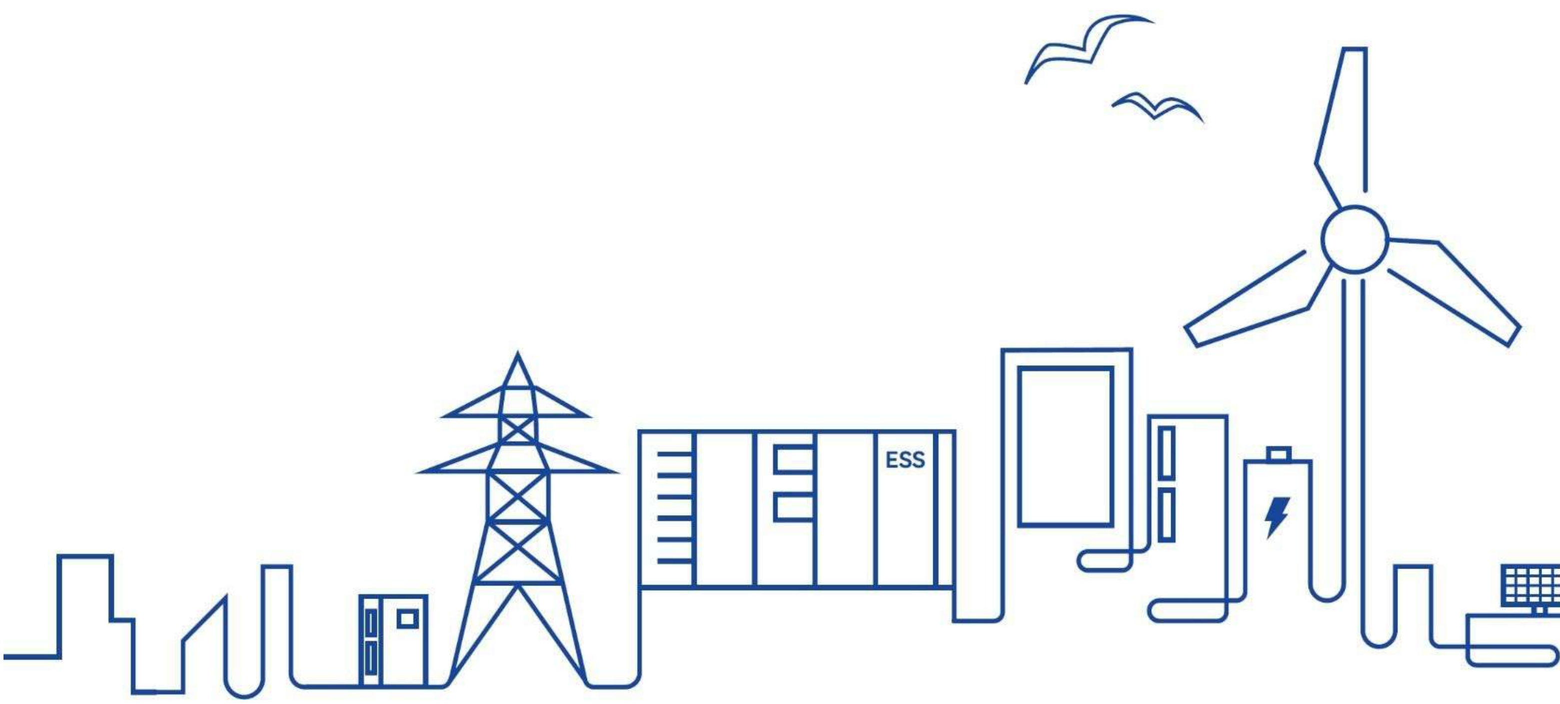

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1. Safety Precautions

1.1 Personnel Requirements

The hoisting, transportation, installation, wiring, operation, and maintenance of the ESS must be carried out by professional electricians in accordance with local regulations. The professional technician is required to meet the following requirements:

- Know electronic, electrical wiring and mechanical expertise, and be familiar with electrical and mechanical schematics.
- Be familiar with the composition and working principles of the ESS and its front- and rear-level equipment.
- Have received professional training related to the installation and commissioning of electrical equipment.
- Be able to quickly respond to hazards or emergencies that occur during installation and commissioning.
- Be familiar with the relevant standards and specifications of the country/region where the project is located.

1.2 Electrical Safety

DANGER

- Touching the power grid or the contact points and terminals in the devices connected to the power grid may lead to electric shock!
- The battery side or the power grid side may generate voltage. Always use a standard voltmeter to ensure that there is no voltage before touching.

DANGER

- Lethal voltages are present inside the device!
- Note and observe the warnings on the product.
- Respect all safety precautions listed in this manual and other pertinent documents.
- Respect the protection requirements and precautions of the lithium battery.

DANGER

- Electricity may still exist in the battery when the power supply of the ESS is disconnected. Wait 5 minutes to ensure the equipment is completely voltage-free before operating.

WARNING

- All operations, such as hoisting, transportation, installation, wiring, operation, and maintenance must comply with the relevant codes and regulations of the region where the project is located.

WARNING

- Always use the product in accordance with the requirements described in this manual. Otherwise, equipment damage may occur.

CAUTION

- To prevent misuse or accidents caused by unrelated personnel, observe the following precautions.
- Post prominent warning signs around the ESS to prevent accidents caused by false switching.
- Place necessary warning signs or barriers near the product.

1.3 Battery Safety

WARNING

Do not leave the product in a low voltage or low SOC condition for a long period of time. Loss of capacity due to the following conditions is not covered by the warranty...

- Battery discharge cell voltage is below 2.8V for 120 consecutive hours.
- Any cell cluster SOC is 0% for 120 consecutive hours.
- Battery discharge cell voltage ≤2.5V.

WARNING

Over or under voltage fault & alarm (detailed information can be found in the Communication protocol).

- Fault: “Cell over voltage fault”, “Cell under voltage fault”, “Total over voltage fault”, “Total under voltage fault”.
- Alarm: “Cell over voltage alarm”, “Cell under voltage alarm”, “Total over voltage alarm”, “Total under voltage alarm”.

End users must assign high priority to above listed faults and alarms reported by the Linyang local controller. When an alarm or fault is triggered, the user interface should prominently highlight these issues. Furthermore, end users should promptly contact Linyang for a timely resolution to prevent battery warranty loss due to over-discharge or overcharge.

CAUTION

- If the system will not be in operation for an extended period (7 days or more), it's recommended to increase the SOC lower limit protection value to above 10% SOC. Additionally, it is important to regularly monitor the system's SOC to avoid the risk of over discharge which will cause warranty expiration.
- During maintenance or shutdown, if the SOC of any battery cluster is 0%, the SOC needs to be charged to 15% and above within 120 hours
- If the SOC of any battery cluster is 0% during operation, the SOC needs to be charged to 5% and above within 2 hours. Or when the SOC reaches 0%, a command can be issued by the host computer EMS to change the system mode to recharge mode.

For safe use of the product, the technician should carefully read and strictly observe the safety requirements. Linyang shall not be liable for product functional abnormality, component damage, personal safety accident, property loss, or other damage caused by the following reasons:

- Batteries are not charged as required, thus resulting in battery capacity loss or irreversible damage.
- Batteries are damaged or dropped, or have leaked, due to improper operations or failure to perform operations as required.
- Batteries are damaged due to over-discharge as they have not been powered on in time.
- Batteries are damaged due to the use of improper equipment for charging and discharging.
- Batteries are frequently over-discharged due to improper maintenance; battery capacity is incorrectly expanded; or batteries have not been fully charged for a long time.
- Battery operation parameters are not correctly set.
- Batteries are damaged because their operating environment does not meet the requirements.
- The customer uses the batteries beyond the scenarios specified in this manual, including but not limited to, connecting extra loads.
- Batteries are not maintained in compliance with the requirements specified in the system manual.
- The product is damaged due to the customer's continued use of batteries beyond the warranty period.
- The product is damaged due to the use of defective or deformed batteries.
- Use the batteries provided by Linyang together with other batteries, including but not limited to batteries of other brands or batteries of different rated capacities.
- Product damage or property loss are caused due to storing or installing batteries together with flammable/explosive materials.
- Personal safety accidents and property loss are caused by battery-related operations performed by non-qualified personnel, or by personnel not wearing qualified protective equipment during operations.
- Batteries are damaged due to improper behaviors, such as eating, drinking, and smoking near the battery.
- Batteries are stolen.

1.4 Hoisting and Transportation

WARNING

- When walking on the top of the equipment, be sure to follow the standard procedure for working at heights.

1.5 Installation and Wiring

WARNING

- In the whole process of mechanical installation, the relevant standards and requirements of the project location must be strictly observed.

WARNING

- Only equipment designated by Linyang can be used. Failure to use equipment designated by LINYANG may cause damage to the protection function and injury to personnel.

1.6 Operation and Maintenance

DANGER

Improper maintenance practices may result in injury to personnel or damage to equipment!

- Make sure that the equipment is disconnected from the power supply before maintenance operations. Use testing equipment to check that there is no voltage or current and wear protective equipment to operate and maintain the equipment.
- After stopping the product, there is still a risk of burns. After the product has cooled down, protective gloves are required before handling the product.

DANGER

- Dismantling or burning the battery may cause it to catch fire.

WARNING

- Personal protective equipment is required for maintenance and service of the ESS. Maintenance personnel must wear protective equipment such goggles, helmets, insulated shoes, gloves, etc.

⚠ WARNING

- There are no user-maintainable parts inside the battery unit.
- Only personnel approved by LINYANG can remove, replace and dispose of the batteries. Users are not allowed to maintain batteries without guidance.

⚠ WARNING

- To avoid electric shock, do not perform any other maintenance operations beyond those described in this manual.
- If necessary, contact LINYANG Customer Service for maintenance.

⚠ WARNING

- To ensure continuous fire protection, replacement of internal components should only be performed by professional personnel.

⚠ WARNING

- Protective tools such as goggles are required when carrying out coolant (glycol solution) or liquid cooling pipeline maintenance.

⚠ CAUTION

- Do not spray paint any internal or external component of the product.
- Do not use cleaning agents to clean the product or expose it to harsh chemicals.

1.7 Personal Protective Equipment

When performing installation, maintenance, and other operations on the equipment, wear personal protective equipment (PPE) that meets the following requirements.

Arc-rated clothing and equipment with an arc rating equal to or greater than the determined incident energy.

- Arc-rated long-sleeve shirt and arc-rated pants or arc-rated coverall or arcflash suit.
- Arc-rated face shield and arc-rated balaclava or arc-rated flash suit hood.
- Arc-rated jacket, parka, or rainwear.

Other PPE

- Hard hat
- Arc-rated hard hat liner
- Safety glasses or safety goggles
- Hearing protection
- Heavy-duty leather gloves or rubber insulating gloves with leather protectors
- Leather footwear

Arc ratings can be for a single layer, such as an arc-rated shirt and pants or a coverall, or for an arc flash suit or a multi-layer system consisting of a combination of arc-rated shirt and pants, coverall, and arc flash suit.

Face shields with a wrap-around guarding to protect the face, chin, forehead, ears, and neck area. For full head and neck protection, use a balaclava or an arc flash hood.

Rubber insulating gloves with leather protectors provide arc flash protection in addition to shock protection. Higher class rubber insulating gloves with leather protectors, due to their increased material thickness, provide increased arc flash protection.

1.8 Product Disposal

When the equipment or its internal components reach the end of life, do not dispose of it together with household wastes. Some components inside the equipment can be recycled, while some may pollute the environment. Contact an authorized local facility for collection.

2. Product Description

2.1 Functions and Features

2.1.1 Product Functions

This container energy storage system is mainly used for storing and transferring energy, which supports battery status monitoring and management, wide application such as grid peak regulation, frequency regulation, grid-connected access of new energy power stations, etc.

2.1.2 Product Features

This Liquid cooling energy storage system is a highly integrated product, which integrates battery system, battery management system, thermal management system, fire suppression system, power supply and distribution system, monitoring system, environmental control system, and cable routing, etc., and has the characteristics of high safety and reliability, easy installation, high energy density and long cycle life.

High energy densit
The large module technology and 1500V high-voltage group technology are used to substantially improve the energy density of the energy storage system, reduce the material cost of single-system equipment and the land cost of the energy storage power station, and improve the economy of the system.

High safety
High-sensitivity temperature, smoke and gas composite detection provide high-reliability protection with the concept of timely detection, timely response, early warning, and prevention of accident expansion. The double-layer fire extinguishing mechanism by Perfluoro + backup water is used to ensure product safety under extreme conditions.

2.2 Product Appearance

Figure 2.1 The dimension of the container system

2.3 System Description

2.3.1 System Specification

Item		Specification
Battery Type		314Ah
Configuration		12P416S
Rated Energy/ kWh		5015
Battery Voltage Range/ V		1164.8 – 1497.6
Duration/ h		h≥2
Communication		Ethernet / CAN / RS485
Thermal Management		Liquid cooling
Dimension (W x D x H)/ mm		6,058 x 2,438 x 2,896
Weight/ T		~ 43
Operation Temperature/ ℃	Charge	0 - 55
	Discharge	-20 - 55
Humidity (storage/operating)		<95% RH (non-condensing)
IP Level		IP54

Table 2-1 Parameters of energy storage system

2.3.2 System Architecture

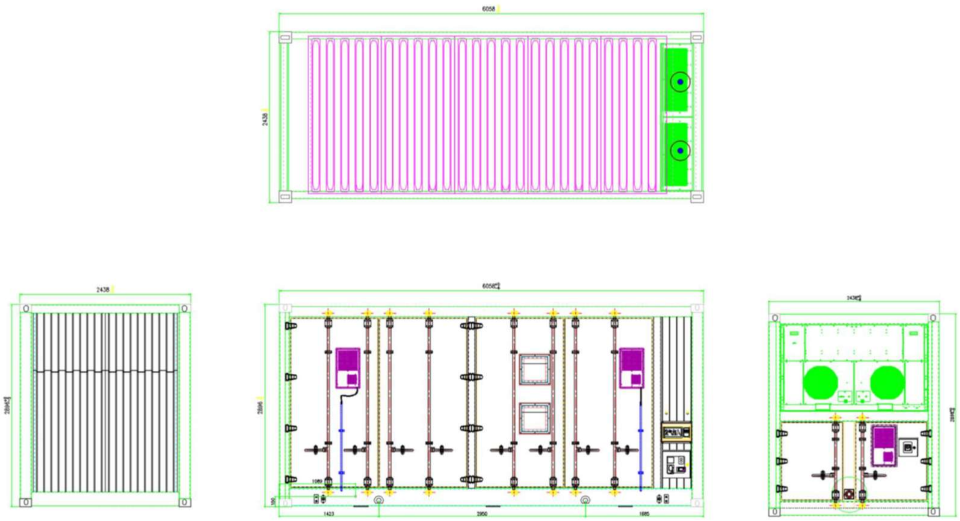


Figure 2.2 The structure diagram of the container system

2.3.3 System Components

No.	Sub-parts	Components	Quantity	Remark
1	Battery Container	20HC	1	Including distribution and lighting system
2	Battery System	Battery pack	48	1P104S
		Battery cluster switchgear unit	12	Including BCMU, breaker, fuse, contactor, current sensor, etc
		DC combiner cabinet	1	Including BAMS, disconnector, UPS
3	Liquid Cooling System	Liquid cooling machine	1	Cooling capacity: 45kW
		Liquid cooling pipes	1	Three-stage pipes

No.	Sub-parts	Components	Quantity	Remark
4	Fire Suppression System	Fire suppression controller	1	Including perfluoro fire extinguishing reagent
		Smoke sensor	3	For smoke detection
		Flammable gas sensor	2	H2
		CO sensor	2	CO
		Exhaust fan	1	
		Ventilation system	1	Flammable gas in battery compartment

Table 2-2 Components of the container system

2.4 Component Description

2.4.1 Battery Cell

Item		Specification	
Battery Type		LFP	
Capacity/ Ah		314	
Nominal Voltage/ V		3.2	
Voltage Range/ V		2.5 - 3.65	
Rated Energy/ Wh		1004.8	
Charge/Discharge Current		Standard	0.5C
		Max	1C
Energy Density/ Wh/ kg		183	
Life Cycle (25℃, 100%DOD, 80%SOH)		6,000 (0.5C/0.5C)	
Dimension (W x D x H)/ mm		72×174×207.7	
Weight/ Kg		5.5	
Operation Temperature/ ℃	Charge	0 - 55	
	Discharge	-20 - 55	

Table 2-3 Parameters of battery cell

2.4.2 Battery Pack

Item		Specification
Pack Type		1P104S
Capacity/ Ah		314
Rated Energy/ Wh		104.499
Configuration		1P104S
Nominal Voltage/ V		332.8
Voltage Range/ V		291.2 – 374.4
Duration/ h		≥2
Operation Temperature/ ℃	Charge	0 - 55
	Discharge	-20 - 55
Dimension (W x D x H)/ mm		762.5x 2180 x 252
Weight/ Kg		~690±10
IP Level		IP67
Thermal Management		Liquid cooling

Table 2-4 Parameters of battery pack

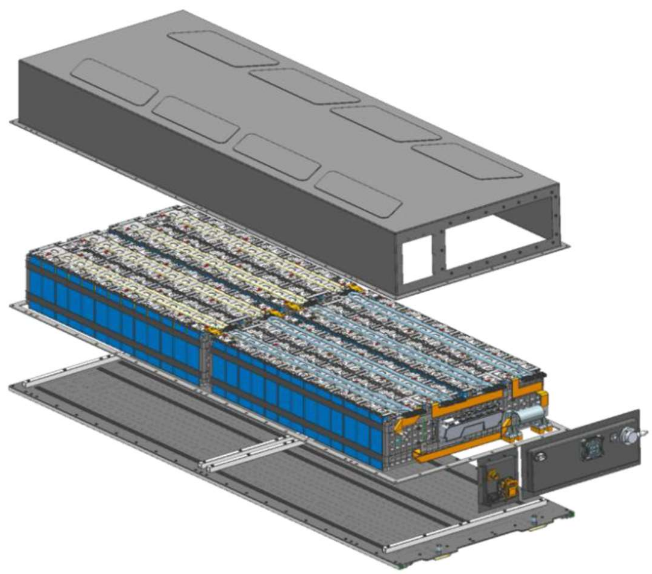


Figure 2-3 The diagram of battery pack (for reference only)

2.4.3 Battery Cluster

Item		Specification
Pack Type		1P416S
Battery Type		LFP
Capacity/ Ah		314
Rated Energy/ Wh		417.996
Configuration		1P416S
Nominal Voltage/ V		1331.2
Voltage Range/ V		1164.8 – 1497.6
Duration/ h		≥2
Operation Temperature/ ℃	Charge	0 - 55
	Discharge	-20 - 55
Weight/ Kg		~ 2.7
IP Level		IP54
Thermal Management		Liquid cooling

Table 2-5 Parameters of battery cluster

2.4.4 Liquid Cooling System

The battery part of the system adopts liquid cooling technology to dissipate heat. Liquid cooling technology brings less loss and better temperature uniformity. The liquid cooling system mainly comprises of pipes, pumps, heat exchangers and compressors. The coolant of the system is mixed solution of ethylene glycol and water. The coolant flows from the water inlet pipe to the 10 main longitudinal branches. Each main branch is divided into 8 branches and flows to the liquid cooling pack to cool / heat the cell. After flowing out of the pack, it is summarized to the return pipe through the branch pipe and returned to the liquid cooling unit.

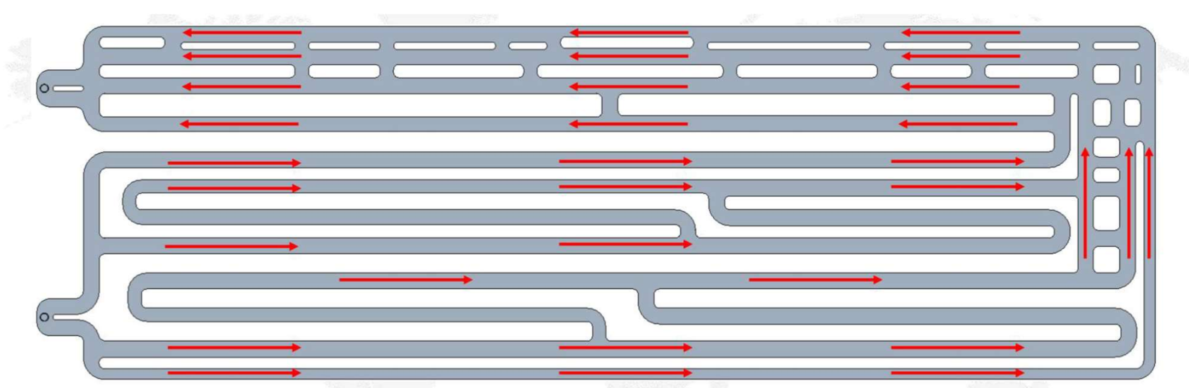


Figure 2-4 Flow direction of coolant in pack

The operating principal diagram of the liquid cooling unit is shown in the figure below.

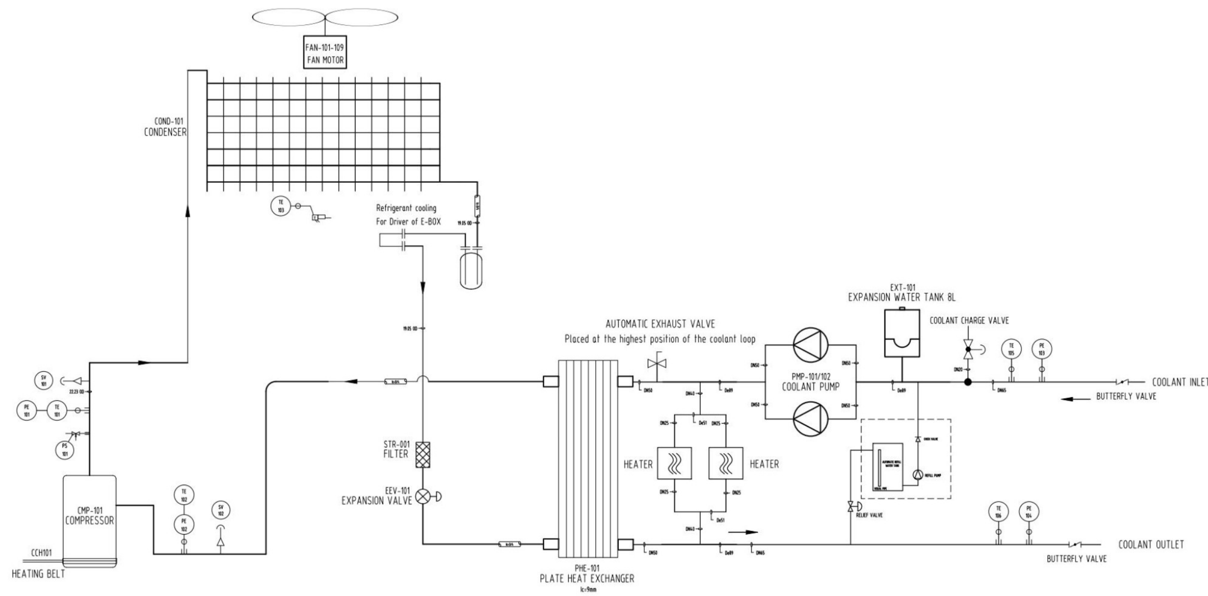


Figure 2-5 Operating principle diagram

Refrigerant Cycle

The compressor compresses the medium temperature and low-pressure gas from the evaporator into high temperature and high-pressure gas. When the gas pass through the condenser, they will liquefy and release heat into medium temperature and high-pressure liquid and then pass through the electronic expansion valve to become a gas-liquid mixed state with low temperature and low pressure (the gas accounts for a small proportion) and finally enter the evaporator. After sufficient heat exchange in the evaporator, the refrigerant turns into a low-temperature and low-pressure gas and then repeat the next cycle.

Coolant Cycle

The coolant is sent to the battery by the pump. After absorbing the heat generated by the battery it is returned to the evaporator to exchange heat, then repeat the next cycle.

2.4.5 Fire Suppression System

Fire protection system introduction

The battery liquid-cooled container system uses a FK5112 fire suppression system and a water-based fire protection system. Each liquid-cooled container is equipped with smoke detectors, heat detectors, combustible gas detectors, and CO detectors, which communicate with the fire control mainframe. The container is also equipped with a fire control mainframe, audible and visual alarms, gas release indicators, and emergency start/stop buttons. Additionally, each battery pack is equipped with an integrated nozzle detection device that can detect changes in CO and temperature within the battery pack, thus transmitting alarm signals to the fire control mainframe for comprehensive judgment.

When any detector inside the compartment is triggered, the fire control mainframe receives a primary alarm, and the audible and visual alarm installed on the container will be activated. When a smoke detector and a heat detector simultaneously trigger an alarm, the fire control mainframe will issue a secondary alarm, the outdoor audible and visual alarm will be activated, the gas fire suppression system will be initiated, and the extinguishing agent will be discharged into the battery compartment. At the same time, the gas release indicator will be activated. The compartment is also equipped with explosion-proof combustible gas detectors and CO detectors. When the detectors reach low or high thresholds, the explosion-proof fans will be activated. Once the concentration of combustible gases drops below the low threshold or the fire protection system reaches a secondary alarm level, the intake and exhaust fans can automatically shut down. The fire control mainframe can issue fire alarm and fault dry contact signals to the battery management system.

When the integrated device in the battery pack reaches a secondary or tertiary alarm level, the fire control mainframe will issue a primary external alarm, which can activate the audible and visual alarm. When it reaches a quaternary alarm level, the fire control mainframe will issue a secondary external alarm, activate the audible and visual alarm, and start a 30-second countdown. At the end of the countdown, the FK5112 system will be activated to spray the extinguishing agent.

Liquid-Cooled container fire suppression process

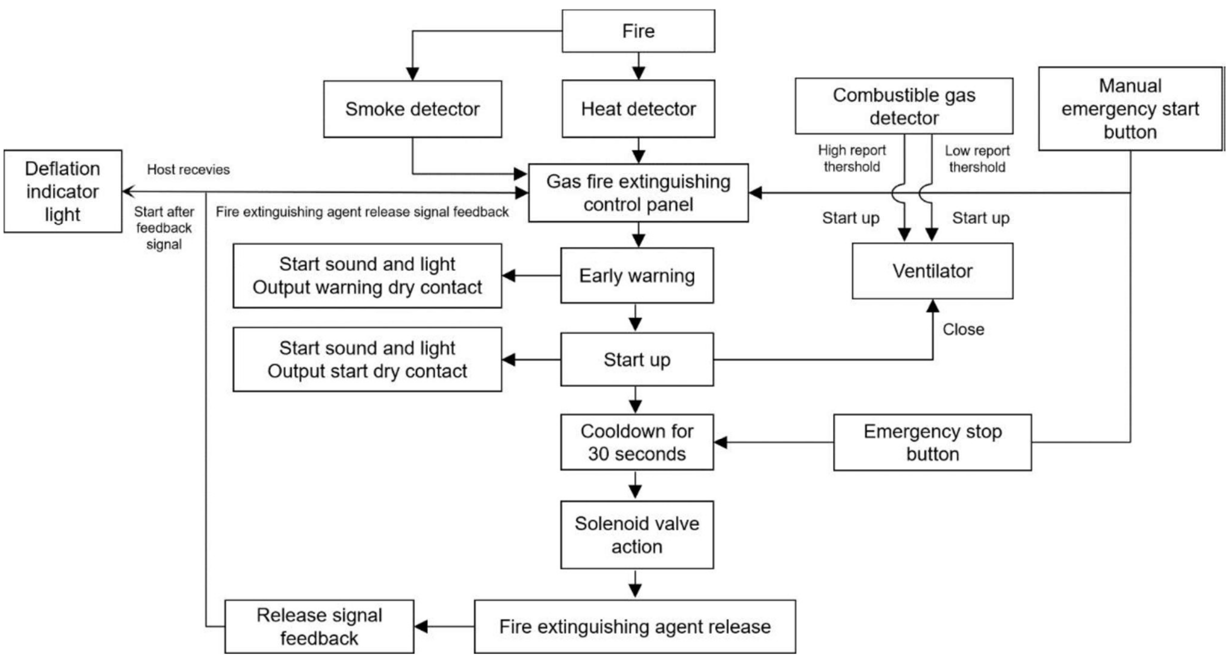


Figure 2-6 Fire alarm process diagram

2.4.6 Battery Management System（BMS）

The battery system uses three level BMS (Battery Management System). It consists of the battery management unit BMU, battery cluster management unit BCMU, and battery stack management unit BAMS. The main functions of BMS are described as follows:

1. Operation control function: including start-stop control, charge and discharge control, operation parameter setting and thermal management control
2. Data acquisition function: BMS should be able to measure the battery's electrical and thermal related data in real time, including single battery voltage, battery module temperature, battery module voltage, series circuit current, insulation resistance, battery pole temperature, leakage monitoring and other parameters.
3. Alarm protection function: BMS has the electrical protection function of battery overvoltage protection, undervoltage protection, overcurrent protection, short circuit protection, overtemperature protection, low temperature protection, leakage protection, etc. and can send alarm signals or trip instructions, implement local fault isolation, and report alarm information.
4. Fault diagnosis function: The BMS should be able to monitor the operating status of the battery in real time, diagnose the abnormal operating status of the battery and the BMS itself, and upload the alarm signal to the local monitoring system and power conversion system.
5. Operation management function: BMS can effectively manage charge and discharge to ensure that no overcharge or over discharge of the battery occurs in the process of charge and discharge, to prevent charging and discharging current and temperature from exceeding the allowable value.

Each battery box is configured with two battery management units (BMUs). The BMUs collect battery data such as voltage and temperature and communicate with the upper-layer main control module through the CAN. The battery cluster is equipped with a control box. The control box integrates the BMS secondary master BCMU and the battery cluster control circuit to manage and control the battery cluster and upload the status information of the battery cluster to the upper management unit. The 18 clusters of batteries in the entire system are controlled and managed by two BMS system management hosts, which upload the status data of the battery system to the energy storage system monitoring device and receive the control instructions of the superior device. The battery system management host also provides display and storage functions.

BMU parameter

No.	Name of technical parameter	Unit	Parameter
1	Number of voltage samples	PCS	52
2	Temperature collection quantity	PCS	12
3	Cell voltage measurement range	V	0~5
4	Cell voltage detection accuracy	mV	±5
5	Cell voltage sampling period	ms	≤100ms
6	Cell temperature detection range	℃	-40~105℃
7	Cell temperature detection accuracy	℃	±1
8	Cell temperature sampling period	ms	≤200ms
9	Battery equalization mode	/	Passive equalization
10	Battery equalizing current	mA	≥80mA
11	BMU power supply	VDC	24

No.	Name of technical parameter	Unit	Parameter
12	BMU power consumption	W	≤2
13	Communication mode	/	CAN2.0

Table 2-6 BMU parameter specifications requirements

BCMU parameter

No.	Name of technical parameter	Unit	Parameter
1	Total voltage measurement range	V	0~1500
2	Total voltage measurement accuracy	%	±0.5%FSR
3	Total voltage measurement period	ms	≤100ms
4	Current measuring range	A	±400
5	Current measurement accuracy	%	±0.5%FSR
6	Current measurement period	ms	≤100ms
7	SOC calculation	/	≤5%
8	Electrical energy calculation error	/	≤±2%
9	Communication mode	/	CAN/RS485/Modbus
10	Communication interface	/	CAN/RS485
11	BCMU power supply	VDC	24
12	BCMU power consumption	W	≤3

Table 2-7 BCMU parameter specifications requirements

BAMS parameter

No.	Name of technical parameter	Unit	Parameter
1	Operating voltage system	VDC	24
2	Operating temperature range	℃	-20~+85
3	Maximum working humidity	%	20%~90%RH
4	Temperature detection range	℃	-20~+105
5	Operating power consumption	W	≤5
6	Communication mode	/	CAN/RS485/Modbus TCP/RTU
7	Communication interface	/	CAN/RS485, ethernet

Table 2-8 BAMS parameter specifications requirements

3. Transport and Storage

3.1 Precautions

CAUTION

Failure to transport and store the product in accordance with the requirements in this manual may invalidate the warranty.

3.2 Transport Methods

ESS can be transported by road and sea. The ESS is highly integrated and easy to hoist, which facilitates its transport. Currently, ESS is not permitted for air transport and there is no specific guidance on rail transport.

ESS leaves its manufacturing factory by truck. While domestic shipments can be made using only trucks, cross-country shipments usually require a combination of truck-ship-trucktransport. In this case, the cargo needs to be transferred from the truck to the ship at or near the port of destination and vice versa.

NOTICE

In most cases, the total weight of the truck and the cargo exceeds the limits allowed by general roads. In such cases, an overweight permit from the country or region of transport may be required.

3.3 Requirements for Transportation

All devices in the ESS have been installed and fixed before leaving the factory, and they can be hoisted and transported as a whole during transportation.

WARNING

In the whole process of loading, unloading, and transportation, the safe operation regulations of ESS in the country/region where the project is located must be observed!

- All tools used for the ESS and during operation shall be properly maintained.
- All personnel engaged in loading, unloading and anchoring should have received relative training, especially in safety.

*During the whole process of loading, unloading and transportation, the mechanical parameters (overall dimensions and weight) of the ESS should always be kept in mind.

The following conditions should be met for the transportation of ESS:

- All container doors are locked.
- Select appropriate crane or lifting tool according to the site conditions. The lifting tool used shall have a sufficient load bearing capacity, boom length and radius of rotation.
- Additional traction may be required if ESS needs to be transported on slopes.
- Remove all obstacles that exist or may exist on the way, such as tree branches, cables, etc.
- The ESS should be transported and moved under good weather conditions.
- Be sure to set up warning signs or warning area to prevent non-staff from entering the lifting area to avoid accidents.

When transporting by road, it is important to fix the bottom corner pieces of the ESS to the transport vehicle to avoid excessive tilt during transportation.

3.4 Storage Requirements

To prevent possible condensation or its bottom from being soaked by rainwater in the rainy season. The ESS should be stored on higher ground.

- Raise the container base if the ESS must be stored outdoors due to site conditions. The specific height should be reasonably determined based on site geological and meteorological conditions.
- Store the ESS on dry, flat, and stable ground with sufficient carrying capacity and without any vegetation cover. The ground must be flat and dry. The upper surface of foundation should be guaranteed to be on the same level, and the level deviation should meet 0~10mm.
- Before storage, ensure that the doors of the container and all internal equipment are locked.
- System storage environment temperature: -30 ℃ ~ 50 ℃, recommended storage temperature: -30 ℃ ~ 25 ℃.
- Long-term storage of batteries is not recommended because it may cause the decrease in battery capacity. Even if the battery is stored at the recommended storage temperature, irreversible capacity fade will still occur during periods of rest. The longer it has been stored, the greater the capacity fade. Please refer to the technical protocol for specific rate of capacity fade.
- The relative humidity should be between 0~95%, without condensation.
- The air inlet and outlet of the ESS should be effectively protected to prevent rain water, sand and dust from penetrating into the container.
- Carry out periodic inspections. Check the container and the inner equipment for damage at least every half a month.
- Before installing a container that has been stored for more than six months, open the door to visually check and ensure that there is no condensation. Check the container and the inner equipment for damage. Check the product after it is powered on and starts. If necessary, request professionals for testing before installation.

- PACKs should be stored in a clean and dry place and not be exposed to the blazing sun and rain. No harmful gases, flammable and explosive products, or corrosive chemicals should be placed at the storage site. Protect the batteries from mechanical shock, heavy pressure, strong magnetic field, and direct sunlight.
- Pay attention to possible hazards in the surrounding environment, such as sudden temperature changes or collisions, to prevent any damage to the PACK.
- Regularly inspect the device. Ensure that the packaging is not damaged in any way and prevent any damage that may be caused by pests and animals. Replace the packaging immediately if it is damaged.
- The packing box cannot be tilted or turned upside down.
- Starting from the date of departure LINYANG factory, the ESS with a storage period of more than 6 months under the above conditions are to be charged and discharged once to bring the system SOC to 30%~40%.

4. Mechanical Installation

WARNING

During the whole process of mechanical installation, the relevant standards and requirements of the project site must be strictly observed.

4.1 Inspection Before Installation

4.1.1 Deliverable Inspection

Check whether deliverable are complete against the attached packing list.

4.1.2 Product Inspection

- Check whether the container received is the ordered one.
- Check the ESS and the internal equipment for any damage.

If any problems are found or there is any question, please contact the forwarding company or LINYANG.

WARNING

- Only install the ESS when it is complete and intact
- Before installation, ensure that
- The ESS is in good condition without any damage
- All internal equipment is in good condition without any damage

NOTICE

Check the equipment for bumped and broken paint, etc. To avoid rusting, paint re-pair is recommended. Please refer to section "Painting Make-up Measures" for specific steps.

4.2 Installation Environment Requirements

4.2.1 Installation Site Requirement

The climate environment and geological conditions, such as stress wave emission and underground water level, should be fully considered when selecting the installation site.

- The environment around the installation site should be dry and well ventilated.
- There should be no trees around the installation site to prevent branches or leaves blown off by heavy winds from blocking the door or air inlet of the energy storage system.
- The installation site should be away from areas where toxic and harmful gases are concentrated, and free from inflammable, explosive and corrosive materials.
- The installation site should be far away from residential areas to avoid noises.

NOTICE

Do not install the device in an environment with vibration and strong electromagnetic field. Strong-magnetic-field environments refer to places

4.2.2 Foundation Requirements

⚠ WARNING

The ESS is heavy as a whole. Before constructing the foundation, it is necessary to inspect the installation site in detail (mainly referring to the geological conditions and environmental climatic conditions, etc.). Commence the design and construction of the foundation only after confirming that all requirements are met.

Unreasonably constructed foundation will bring great troubles to the installation of the ESS, affecting the normal opening and closing of the doors and the normal operation. Therefore, the foundation of the ESS must be designed and constructed according to certain standards to meet the requirements of mechanical support, cable routing and later maintenance and overhaul.

At least the following requirements shall be met during foundation construction:

- The soil at the installation site should be compact. It is recommended that the relative density of soil at the installation site be no less than 98%. Take relevant measures to ensure a stable foundation in case of loose soil.
- Compact and fill the foundation pit to provide sufficient and effective support for the container.
- The upper surface of ESS foundation should be guaranteed to be on the same level, and the level deviation should meet 0~10mm.

NOTICE

- If the leveling deviation exceeds the required level, the machine must be leveled by welding pads before it is lifted onto the foundation.
- Once the machine has been installed in position, it should be ensured that the door panels can be opened and closed smoothly before the machine is secured to the foundation.

- The height of the battery container placed off the ground should be implemented according to the requirements of the local design specifications. It is suggested that the height of the foundation should be higher than the highest water level in history, to prevent the container base and the interior from rain erosion.
- The cross-sectional area and height of the foundation should meet the requirements.
- Construct corresponding drainage in conjunction with local geological conditions. To prevent the bottom or internal equipment of the ESS from being soaked in water during the rainy season or during heavy rainfall.
- Built a cement foundation with sufficient cross-sectional area and height. The foundation height is determined by the construction party according to the site geology.
- Consider cable routing when building the foundation.
  * To facilitate subsequent electrical wiring, it is recommended to preset a cable trench in the foundation according to the position of cable inlet holes of the ESS and pre-embed the conduit.
  *The dregs excavated during foundation construction should be removed immediately to avoid latter impact on lifting.
- Sufficient clearance should be maintained around the device for the ease of door
- opening. At the same time, pay attention to the space distance required by the air inlet
- and outlet. For specific dimensions, please refer to “The diagram of the required
- space when the door is opened”.
- If it is fixed by welding, channel steel needs to be embedded. And do a good job of corrosion treatment.
- During the foundation construction, reserve enough space for the AC/DC side cable trench according to the position and size of the cable inlet and outlet holes of the ESS and preempted the cable conduit.
- Determine the specifications and quantity of the perforating gun according to the model and quantity of the cables.

- Both ends of all embedded pipes should be temporarily sealed to prevent impurities from entering and causing trouble to later wiring.
- After all cables are connected, cable inlet and outlet and connector should be sealed with fireproof mud or other suitable materials to prevent rodent access.

* Preempted the grounding unit according to the relevant standards of the country/region where the project is located.

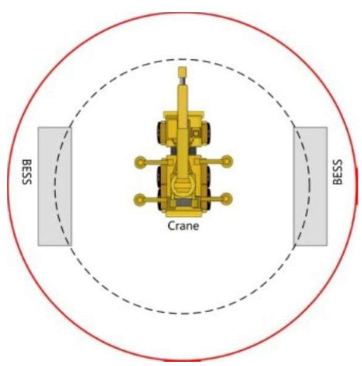
4.2.3 Foundation Leveling Program

The flatness of the foundation should be measured and accepted before the equipment is lifted and fixed. The equipment is installed on foundation, and the position are distributed as shown in the figure below.

Figure 4-1 Foundation design reference plan (top view)

* The figure is for reference only. The product received may differ.

If the height difference between the highest point and the lowest point of 1~10# foundation is within the range of 0~10mm, then the foundation flatness meets the design requirements, and the equipment can be lifted and fixed with the foundation normally; if the height difference is more than 10mm, then it is necessary to carry out leveling of the foundation at the site.



4.3 Hoisting and Fixing

4.3.1 Lifting Precautions

WARNING

- In the process of lifting, it is necessary to operate in strict accordance with the safety operation rules of the crane.
- No one is allowed to stay within 5m to 10m of the operating area. In particular, it is strictly prohibited to stand under the lifting arm and the lifted machine to avoid casualties.
- In case of bad weather, such as heavy rain, fog, gust, etc., the lifting work should be stopped.

When lifting the ESS, ensure that at least the following requirements are met:

- Lift from the top lifting holes and ensure on-site safety during lifting.
- Professional personnel should direct the whole lifting process on site.
- Select appropriate lifting machine according to the site conditions. It is recommended that the bearing capacity of the selected lifting machine shall ≥ 100,000kg.
- The strength of the sling used should be able to bear the weight of the ESS.
- Ensure safe and reliable connections of all slings and an equal length of slings connected to corner fittings.
- The sling length can be adjusted according to the actual situation on site.
- Ensure that the ESS is steady and not tilting during lifting.
- Take all necessary auxiliary measures to ensure safe and smooth lifting of the ESS.

How the ESS is hoisted by a crane is shown in the figure below. The inner dashed circle indicates the crane operation range. When the crane is working, it is strictly forbidden to stand in the solid circle!

4.3.2 Lifting

Lift the ESS according to the following requirements:

- The ESS should be lifted vertically. Never drag the container on the ground or on the top of the lower container, and never pull and push it on any surface.
- Lift the ESS slowly. And during lifting, theoretically, it is required to ensure that the center of the hanger and the center of the ESS top is exactly right. In practice, try to minimize the deviation of the two centers, and ensure that the hanger and the ESS top is parallel through visual inspection to ensure the stability of the lifting. The crane should move at a very slow speed at the moment of lifting and lift at a constant speed later.
- When the ESS is in place, place it lightly and smoothly. It is strictly forbidden to throw it to places outside the vertical landing place.
- The ESS should be placed on a solid and flat site with good drainage and no obstacles or protrusions.

Please choose suitable crane and wire rope according to the actual weight of container.

Lifting instructions

- Container size :6058mm*2438mm*2896mm.
- When hoisting, please confirm that there is no obstacle around, slowly and steadily hoisting, to ensure hoisting safety.
- Do not scratch the container paint when installing the lifting wire rope.
- Lifting speed≤5m/min

*The figure is for reference only. The product received may differ.

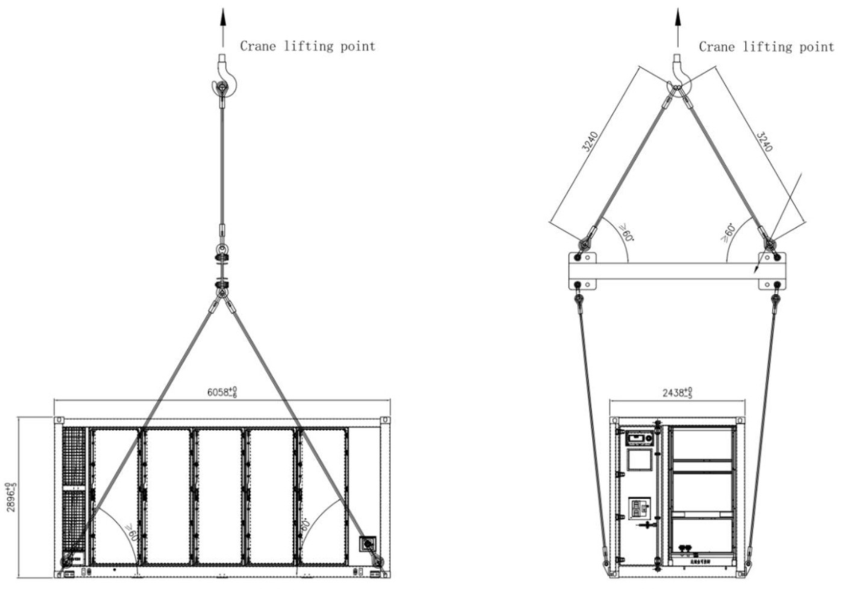


Figure 4-2 lifting form bottom corner casting

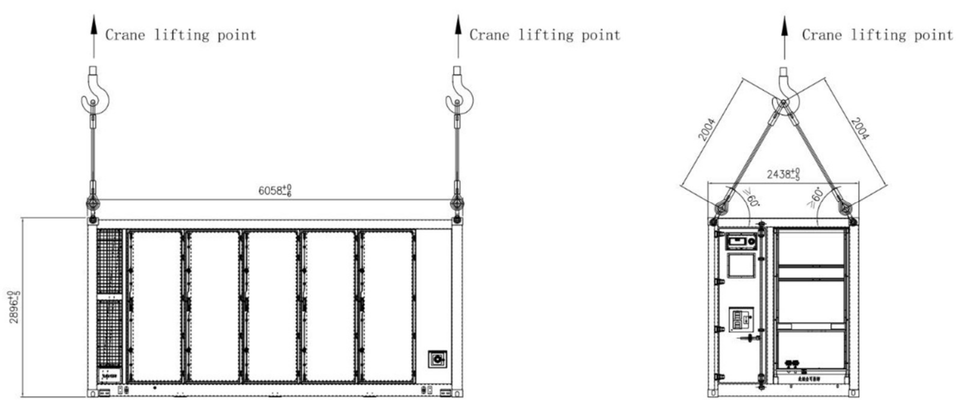


Figure 4-2 lifting form top corner casting

4.3.3 Fixed Installation

WARNING

- The lifting work shall be in accordance with the relevant standards and specifications of the country/region where the project is located.
- LINYANG shall not be held liable for any personal injury or property damage caused by violating relevant requirements or other safety precautions.

The ESS shall be fixed after being transported to the installation position.

Bottom Fixation

Welding fixation method

The container is installed and fixed in a way that is based on the site conditions. The bottom of the container must have a concrete foundation of sufficient strength. When installing the container, the four corners and the bottom side beams of the container must have sufficient support. Embedded channel steels, weld and fix the bottom of the container with the foundation. After completion, anti-corrosion treatment shall be taken for the welding position.

All welding components require to be fully welded, and after the welding is completed, anti-corrosion treatment is applied to the welds.

The container is fixed by welding, which is to weld the four corner pieces at the bottom of the container directly to the embedded steel plate. The fixing method is shown in the figure below:

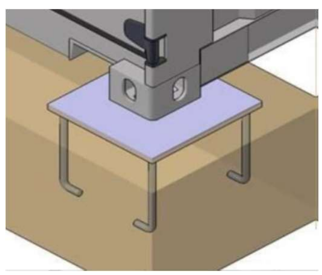
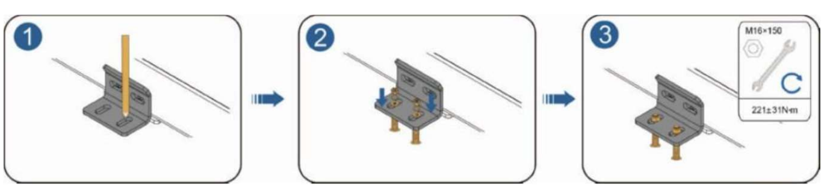


Figure 4-4 Schematic diagram of container welding and fixing

* The figure is for reference only. The product received may differ.

Fixed by L-shaped angle steels

Positions need to be fixed with L-shaped angle steels at the bottom of the ESS are circled in the figure below.



5. Electrical Connection

5.1 Precautions

DANGER

High voltage! Electric shock!

- It is strictly forbidden to directly touch the live parts in the unprotected state!
- Before installation, ensure that the all switches are off.

WARNING

Sand and moisture penetration may damage the electrical equipment in the ESS or affect their operating performance!

- Avoid electrical connections during sandstorms or when the relative humidity in the surrounding environment is greater than 95%.
- Perform electrical connection when there is no sandstorm and the weather is fair and dry.

WARNING

- Before wiring, check and ensure that the polarity of all input cables is correct.
- During electrical installation, do not forcibly pull any wires or cables, as this may compromise the insulation performance.
- Ensure that all cables and wires have sufficient space for any bends.
- Adopt the necessary auxiliary measures to reduce the stress applied to cables and wires.
- After completing each connection, carefully check and ensure that the connection is correct and secure.

WARNING

When an external short circuit occurs in the Rack circuit, or when the terminal box fuse generates a protective action, the fuse in the terminal box and the inter-cluster fuse must be replaced, and the fuse in the MSD must be inspected for damage before use.

5.2 Overview of Wiring Area

The wiring diagram of the integrated ESS is shown below:

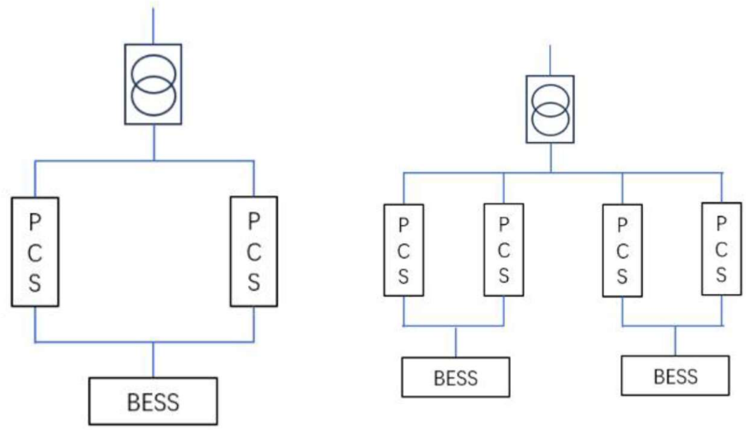


Figure 5-1 Wiring diagram

*The diagram only described the on-site wiring architecture, and the internal wiring is for reference only.

No.	Description	Recommended cable specifications
1	Main power wiring port	240 mm2
2	Communication port	CAT6
3	Auxiliary power supply port	4*35mm+1*16mm2
4	FSS communication port	STP
5	Grounding point	120 mm2 ~ 150 mm2 yellow and green cable or grounding flat steel

5.3 Preparation Before Wiring

NOTICE

- The installation scheme of the ESS must be in full accordance with the regulations or standards where the project is located.
- Failure to follow the installation requirements in this manual may result in faulty device or system, and the damage caused is not covered by the warranty.

5.3.1 Preparing Installation Tools

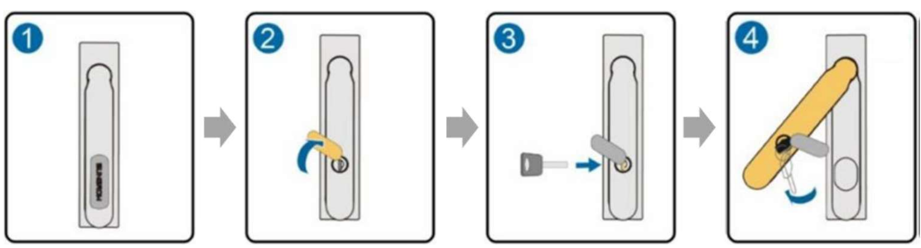
WARNING

- All electrical connection must be carried out strictly in accordance with the wiring diagram.
- All electrical connections must be carried out when the equipment is completely uncharged.

WARNING

Only qualified electricians can perform the electrical connection. Please comply with the requirements in "Safety Precautions" in this manual. LINYANG shall not be held liable for any personal injury or property damage caused by ignoring these safety precautions.

5.3.2 Open the Door



5.3.3 Prepare Cables

The cables must meet the following requirements:

- The current carrying capacity of the cable meets requirements. Factors affecting the current carrying capacity of a conductor include but are not limited to:
  - Environmental conditions
  - Type of the insulation material of the conductor
  - Cabling method
  - Material and cross-sectional area of the cable
- Select cables with a proper diameter according to the maximum load, and the cables should be long enough.
- All DC input cables must be of the same specifications and materials.
- AC output cables of three phases must be of the same specifications and materials.
- Only flame-retardant cables can be used.

NOTICE

- The cables used should comply with requirements of local laws and regulations.
- The cable color in figures in this manual is for reference only. Please select cables according to local standards.

5.3.4 Copper Wire Connection

When copper cables are selected, the connection sequence of wiring parts is shown in the following figure.

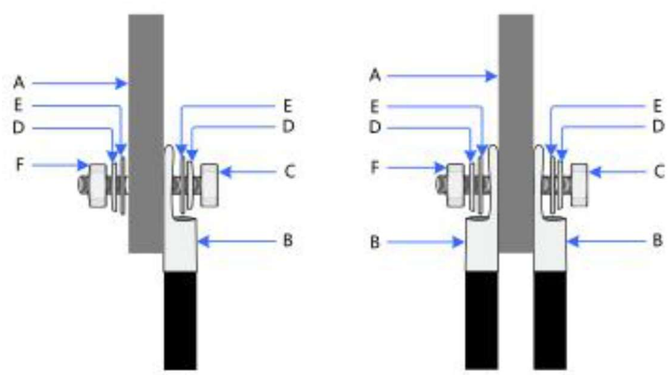


Figure5-4 Copper cable terminal connection

Position	Designation
A	Copper bus bars
B	Copper connection terminal
C	Bolt
D	Spring washer
E	Flat washer
F	Nut

NOTICE

Bolt fastening should be firm and reliable, and the exposed wire buckle should not be less than 2 buckles.

5.4 Ground Connection

The container provides users with at least 2 grounding points in the form of grounding bar bolt connection to achieve reliable grounding of the non-functional conductive conductor of the entire container. The effective cross-sectional area in the grounding system is not less than 250mm².In order to achieve a good grounding effect and realize safe and reliable operation of the system, the grounding impedance of the grounding point provided by the user must be ≤4Ω and the connection impedance must be ≤0.1Ω.

There is a grounding bar inside the container, and the grounding wires of the distribution cabinet, control cabinet, etc. are connected to the internal grounding bar.

The front end of the container and one side wall are each equipped with a lightning protection grounding point, which is connected to the iron bar pre-buried in the foundation at the installation site, as shown in the figure below.

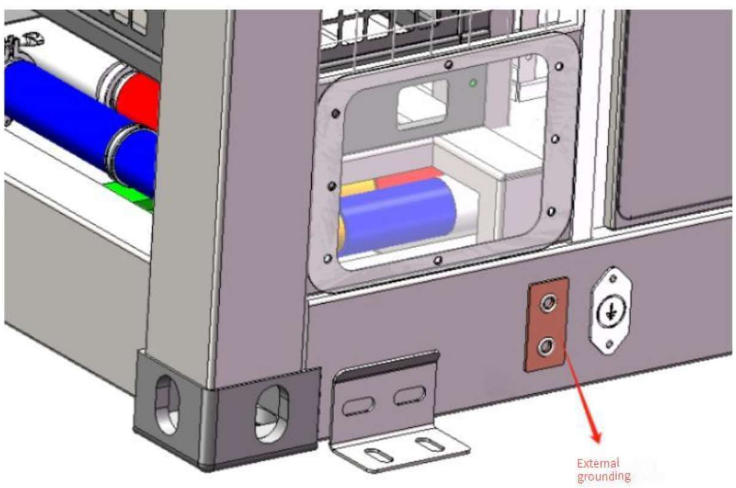


Figure 5-6 Container grounding point diagram

*The figure is for reference only and the actual product shall prevail.

WARNING

- After the completion of the grounding operation need to be in addition to the grounding fixed point of the bare metal layer of good corrosion protection.
- Please perform the external grounding connection according to the actual on-site condition and the instructions of the plant personnel.
- During the grounding connection process, please contact the relevant staff if you have any questions. Failure to follow the installation specifications, or private installation or modification may lead to safety accidents or equipment damage. LINYANG is not liable for any damage caused by this.

5.5 Post-wiring Operations

Check the wiring thoroughly and carefully when all electrical connections have been completed. In addition, perform the following operations:

- Check all air inlets and outlets for blockage.
- Seal the gap around the cable inlet holes.
- Put all protective covers back in place firmly.

WARNING

- Moisture may enter the product if it is not properly sealed.
- Rodents may enter if the product is not properly sealed.

Locking Container Doors

Step 1 Reinstall the protection cover of the wiring area in the reverse order of removal.

Step 2 Lock the container door, pull out the key, and store it securely.

NOTICE

- In order to avoid the door lock latch and door frame hinges bump, it is recommended that the door handle and vertical direction angle is not less than 100° when the door lock is closed.
- Make sure that the seal around the container door does not curl when the door is closed!

6. Battery Connection

6.1 Precautions

Always follow the safety instructions in this manual. In order to avoid personal injury and property damage that may occur during installation or operation, and extend the service life of this product, please carefully read all safety instructions.

Improper or incorrect use may result in:

- A threat to the life and personal safety of the operator or third parties
- Damage to the energy storage system or other property of the operator or third party

* The safety precautions in this manual do not cover all specifications to be followed, and all operations should be performed based on the site conditions.

*LINYANG shall not be liable for any loss arising from failure to follow the safety precautions in the manual.

WARNING

- While installing the device with hazardous voltage, follow relevant regulations and local installation safety guidelines.
- Please observe the regulations on the correct use of tools and personal protective equipment.
- All connections must be carried out with distinctive guidance. Any guess and ambiguous attempts must be prohibited.
- Tools with an insulating protective coating must be used.

Connecting cables should meet the voltage and current requirements.

- All connectors must be safe and reliable to avoid loosening or virtual contact. They
- must be corrosion-resistant, wear-resistant and shock-proof.
- All connections must comply with the requirements of relevant national standards to pre-vent arc discharge in any form.
- The connections of internal batteries must be equipped with anti-vibration and anti-loosening devices. Temperature, voltage and current sensors must be connected safely and reliably, to prevent loosening, ageing and extrusion. All sensor cables must be free of metal exposure.
- Any type of short circuit should be prevented in the connection process.
- Operators must use this product with personal protective equipment.
- Key connections must be correct, reliable (without loosening) and in good contact, without short circuits.
- All the finished connections must be measured and confirmed one by one.

- All connections must not be in contact with the casing or other components or short-circuited.

If there are other uncertain factors, please consult the after-sales technicians of LINYANG before any operation

6.2 Cable Connection

Power cables and wiring harness types:

The high-voltage wiring harness used in the system includes the high-voltage wiring harness inside the cluster and the high-voltage wiring harness outside the cluster. The high-voltage wiring harness inside the cluster connects the battery plug-in boxes in series. Both ends of the high-voltage wiring harness of each battery plug-in box use quick-plug connectors, which have a fool-proof function. The battery cluster is connected to the busbar through the high-voltage wiring harness outside the cluster. One end of the high-voltage wiring harness outside the cluster uses a quick-plug connector, and the other end uses an M10 copper tube terminal to connect to the busbar.

Figure 6-1 Wiring harness between high voltage boxes

Figure 6-2 High voltage harness within the cluster

WARNING

When the plug is plugged tightly, it will make a "click" sound. Make sure the plug is plugged in tightly!

7. Powering up and Shutdown

7.1 Power-on Operation

WARNING

The ESS can only be put into operation after confirmation by a professional and approved by the local power department.

WARNING

For ESS with a long shutdown time, check the equipment thoroughly and carefully to ensure all indexes are acceptable before powering it on.

7.1.1 Inspection Before Powering up

- Before powering up the equipment, check the following items carefully.
- Check whether the wiring is correct.
- Check whether the protective covers inside the equipment are installed firmly.
- Check whether the emergency stop button is released.
- Check and ensure that there is no grounding fault.
- Check whether the AC and DC voltages meet startup conditions and ensure that there is no over-voltage with a multimeter.
- Check and ensure that no tools or components are left inside the equipment.
- Check all air inlets and outlets for blockage.
- If the ESS has been stored for more than six months, the Liquid cooling system fan should be checked for proper rotation, noise or stalling before powering up.

8. Routine Maintenance

8.1 Precautions Before Maintenance

WARNING

- Do not open the door to maintain the device in rainy, humid or windy days. LINYANG shall not be held liable for any damage caused by violation of the warning.
- Avoid opening the container door when the humidity is high in rain, snow or fog, and make sure that the seals around the container door do not curl when the door is closed.
- Before performing maintenance on a Rack, make sure that all MSDs in that Rack have been properly disconnected.
- When performing maintenance on the Pack, ensure that the locking structure of the plug is in the unlocked state before unplugging the power cord to avoid damage to the Pack caused by strong unplugging.

WARNING

- To avoid electric shock, do not perform any other maintenance operations beyond this manual.
- If necessary, contact LINYANG customer service for maintenance.

NOTICE

- In the event of heavy snowfall at the project site, please clear the snow from the top of the equipment and the surrounding area in a timely manner.

8.2 Maintenance Item and Interval

This section is the recommended maintenance cycle. The actual maintenance cycle should be adjusted according to the specific installation environment of this product.

The power station scale, installation location and on-site environment affect the maintenance cycle of this product. In sandy or dusty environments, it is necessary to shorten the maintenance cycle and increase the frequency of maintenance.

8.2.1 Maintenance (Initial grid connection)

Check the following items and correct them immediately if they do not meet the requirements:

- Examine the input and output cable materials and specifications.
- Terminal materials, specifications and their installation direction.
- Bolt size and its gasket mounting direction.

8.2.2 Maintenance (Every month)

Item	Check method
Container	• Check whether there is any damage, flaking paint or sign of oxidization on the enclosure. • Check whether there is any damage or deformation of the container and internal devices • Check whether there are flammable objects on the top of the container. • Check whether the welding points between the container and the • foundation steel plate are firm and whether there is corrosion. • Check whether the lock of the container door can be unlocked flexibly. • Check whether the sealing strip is fixed properly. • Check whether there are foreign objects, dust, dirt, and condensed water inside the integrated energy storage system.
Air inlet and outlet Cables	• Check to see if the outdoor container air inlet and outlet are blocked. • Check all cables for breakage.
System status	• Check if there is abnormal noise during operation of internal devices. • Check whether the temperature in the container is excessively high. • Check whether the humidity and the amount of dust inside the container are within the normal range. Clean the equipment if necessary.
Software data	• Read the data from BMS • Save run data, parameters, and logs to the relevant files. • Check the various parameter settings. • Update the software.

8.2.3 Maintenance (Every half a year)

Item	Check method
Safety function	• Check whether the emergency stop button work normally. • Simulate shutdown. • Check the warning marks and other device marks and replace them timely when they are fuzzy or damaged.
Internal components inspection	• Check the cleanness of the circuit board and other elements and components. • Check to see if the fan is operating normally and if there is any abnormal noise. • Check the temperature of the radiator and the amount of dust accumulated. Clean heat-dissipation modules with a vacuum cleaner if necessary. • Replace the air filter screen when necessary.

Item	Check method
Device maintenance	• Carry out regular inspection for corrosion of all metal components (once per half a year). • Check the contactors (auxiliary switches and micro-switches) annually to ensure the good mechanical operation. • Check the running parameters (especially voltage and insulation). • Troubleshoot for un-operated UPSs, which need to be recharged once every six months.

8.2.4 Maintenance (Once a year)

Item	Check method
Ground of the shielded layer of cables	• Check whether the cable shielding layer is in good contact with the insulation sleeve and whether the copper bus bar is firmly fixed.
Surge protection device and fuse	• Check whether the SPD and fuse are properly fastened.
Wiring and cable layout	• Check whether the cable layout is normal and whether there is a short circuit. For any non-conformances found during inspection, correct them immediately. • Check whether all cable entry are well sealed. • Check whether the power cables are loose and fasten them again by the torque specified previously. • Check whether the power cables and control cables are damaged, especially if the surface contacting the metal surface is cut. • Check whether the insulation tapes on the power cable terminals fall off.
Ground connection and equip connection	• Check whether the ground connection is correct and the grounding resistance shall be no more than 4Ω. • Check whether the equip connection inside the integrated system is correct.

9. Contact Information

In case of questions about this product, please contact us.

We need the following information to provide You the best assistance:

- Model of the device
- Serial number of the device
- Fault code/name
- Brief description of the problem

For detailed contact information, please connect: 8631211